LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Principles of pressure refueling and de-fueling
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Principles of pressure refueling and de-fueling
5. Specific Objectives	17.1.1.1 Trainee will be able to understand the construction and requirements of typical aircraft fuel system and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of fuel system diagrams; Instructor supervision of theoretical concepts only
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the aircraft fuel system?
(b) Name two methods used for refueling aircraft.
(c) Why is pressure refueling preferred over gravity refueling for large aircraft?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and working principle of pressure refueling • Fuel tank pressure requirements and limits • Refueling nozzle design and sealing mechanisms • Fuel flow control valves and shut-off systems • Safety protocols during pressure refueling operations	L & Ds	PPT, IB	15 min
(b)	• Principles of de-fueling and defueling procedures • Pressure differential management during de-fueling • Fuel transfer pumps and venting systems • Precautions against fuel spillage and contamination • Emergency de-fueling scenarios and response	L & Ds	PPT, PDF	20 min
(c)	• Comparison of pressure refueling and de-fueling systems • System components specific to pressure operations • Maintenance requirements for refueling equipment • Regulatory standards for fueling operations • Common failures and troubleshooting techniques	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of pressure differential in preventing fuel tank over-pressurization.
(b) Describe the sequence of operations during a pressure refueling event.
(c) Identify three critical safety measures required during de-fueling operations.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of pressure refueling and de-fueling discussed in this lesson will assist in understanding fuel contamination issues covered in LP 06.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Fuel contamination with water, foreign particles, other types of fuel, micro biological growth, contamination detection and control	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min